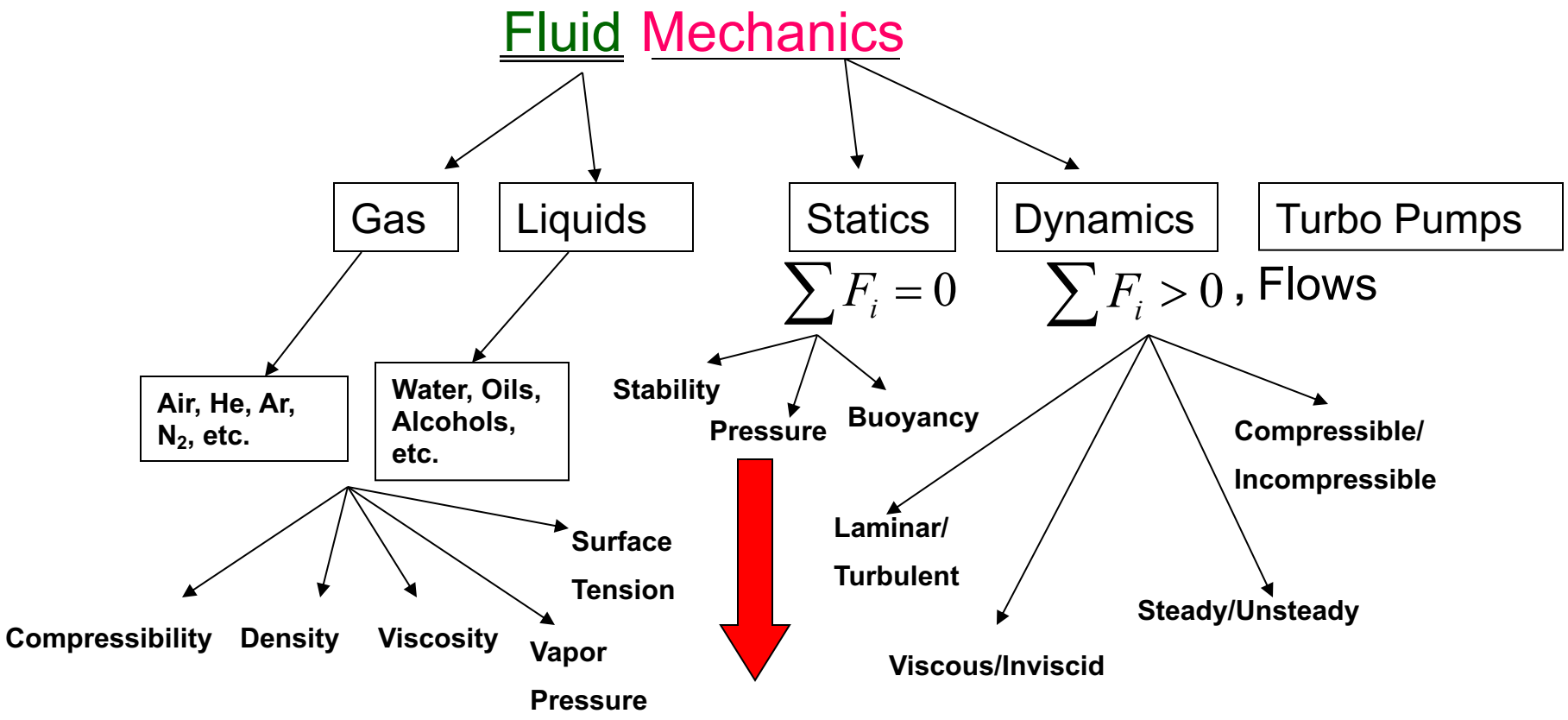
FLUID MECHANICS FOR AEROSPACE ENGINEERING

By

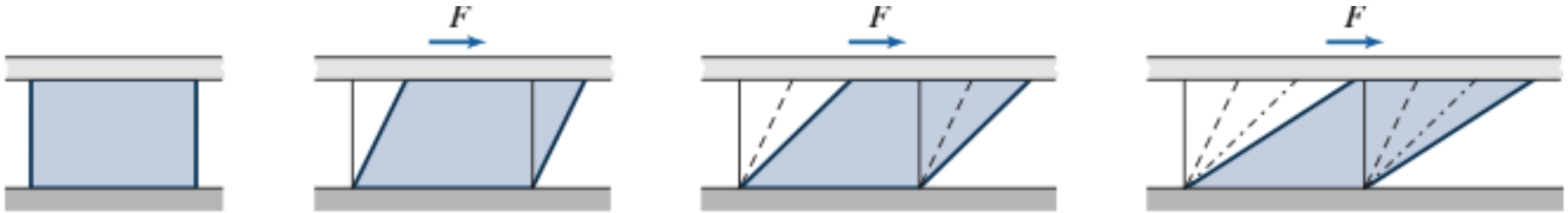
Dr. Ankush D Tharkar

Fluid Mechanics Overview



Fluid Statics

- Fluids tend to flow when we interact with them while solids tend to deform or bend

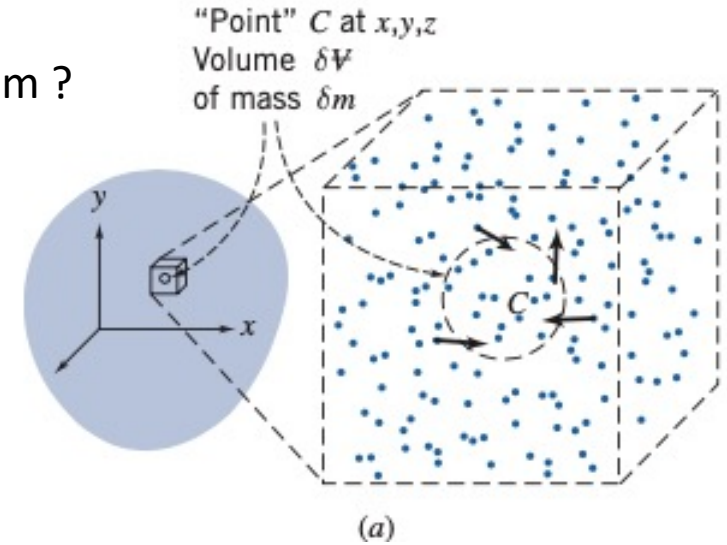


(a) Solid or fluid (b) Solid or fluid (c) Fluid only (d) Fluid only

- A fluid is a substance that deforms continuously under the application of a shear (tangential) stress no matter how small the shear stress may be.
- fluid as any substance that cannot sustain a shear stress when at rest.
- The amount of deformation of the solid depends on the solid's modulus of rigidity G; while learn that the rate of deformation of the fluid depends on the fluid's viscosity μ.

Fluid as a Continuum

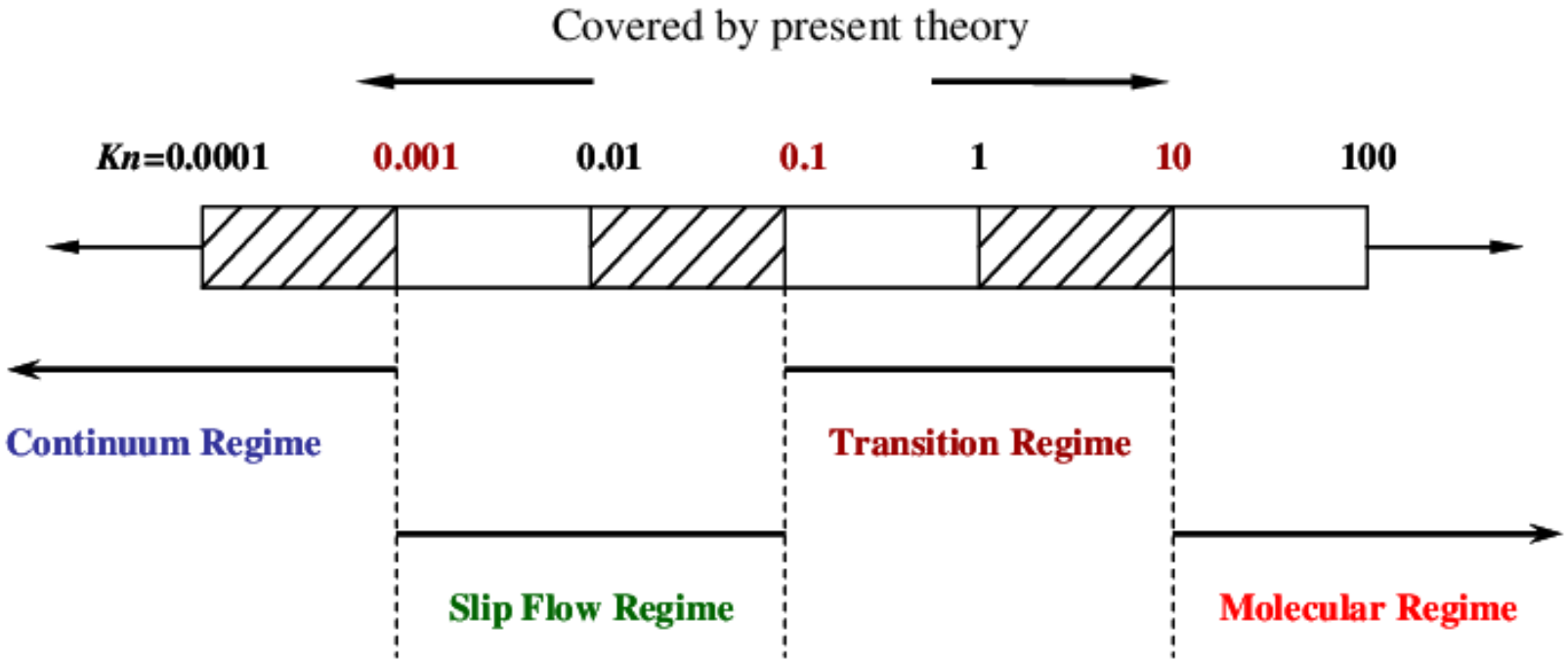
- This molecular structure is one in which the mass is not continuously distributed in space, but is concentrated in molecules that are separated by relatively large regions of empty space.
- A region of space “filled” by a stationary fluid (e.g., air, treated as a single gas) looks like a continuous medium, but if we zoom in on a very small cube of it, we can see that we mostly have empty space, with gas molecules scattered around, moving at high speed.
- What is the minimum volume, δV, that a “point” C must be, so that we can talk about continuous fluid properties such as the density at a point?
- In other words, under what circumstances can a fluid be treated as a continuum ?



- The mean free path is the average distance traveled by a moving molecule between collisions.
- It only breaks down when the mean free path of the molecules becomes the same order of magnitude as the smallest significant characteristic dimension.
- This occurs in such specialized problems as rarefied gas flow (e.g., as encountered in flights into the upper reaches of the atmosphere). For these specialized cases (not covered in this text) we must abandon the concept of a continuum in favour of the microscopic and statistical points of view.
- Characterized by Knudsen Number
- $Kn = \frac{Mean\ Free\ Path}{Characteristic\ Dimension}$
- According to the magnitude of *Kn*, the gas flow regimes are classified as : continuum flow ($Kn < 0.001$), slip flow ($0.001 < Kn < 0.1$), transition flow ($0.1 < Kn < 10$), and free molecular flow ($Kn > 10$).

- The continuum assumption is valid in treating the behaviour of fluids under normal conditions.

- Density: As gas density increases, the molecules become closer to each other. Therefore, they are more likely to run into each other, so the mean free path decreases.
  - Increasing the number of molecules or decreasing the volume causes density to increase. This decreases the mean free path.

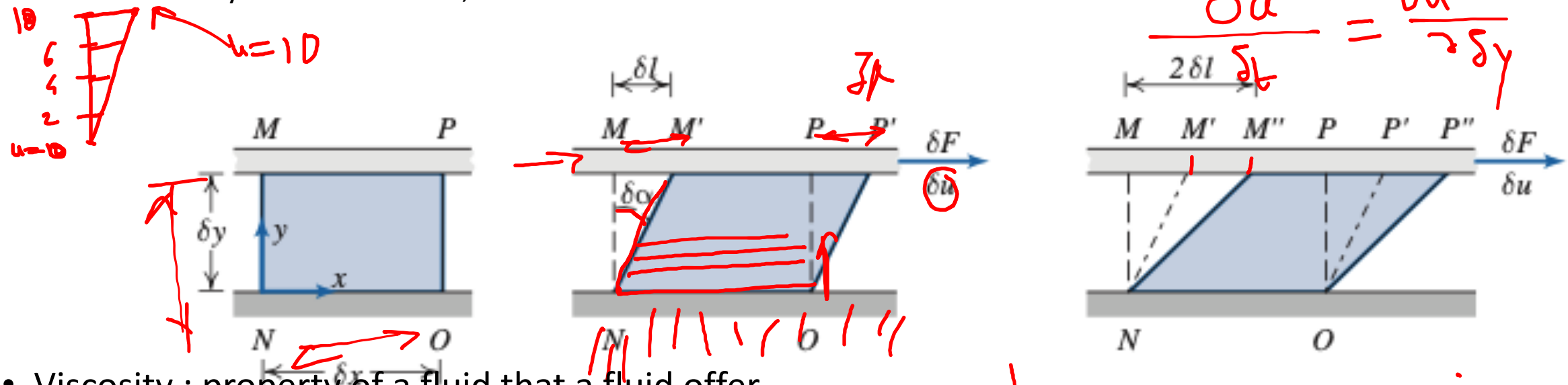


Fluid Properties

- Density
- Viscocity
- Kinematic Viscocity
- Surface Tension
- Surface Energy

Viscosity

- Where do stresses come from?
- For a solid, stresses develop when the material is elastically deformed or strained;
- for a fluid, shear stresses arise due to viscous flow (we will discuss a fluid's normal stresses shortly). Hence we say solids are elastic, and fluids are viscous



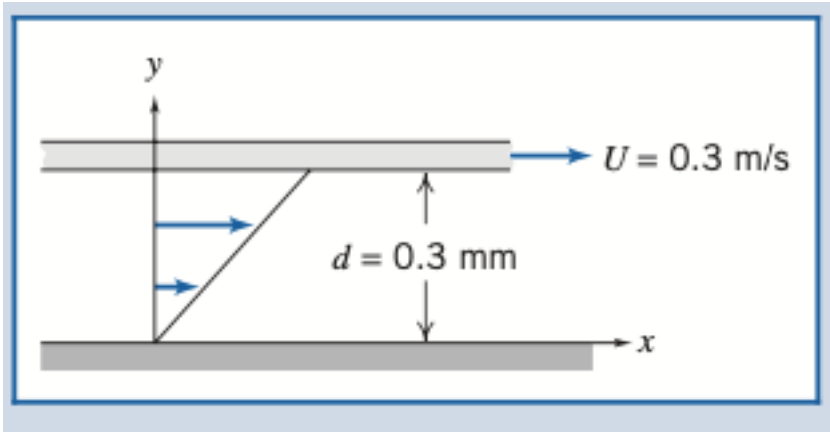
- Viscosity : property of a fluid that a fluid offer resistance to shear stresses
- It arises from intermolecular cohesion and transfer of intermolecular momentum.

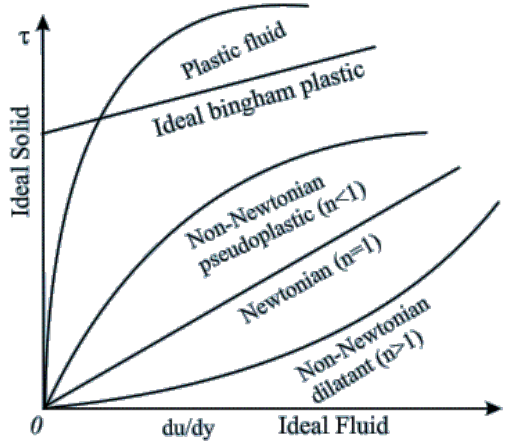
- An infinite plate is moved over a second plate on a layer of liquid as shown. For small gap width, d, we assume a linear velocity distribution in the liquid. The liquid viscosity is 0.65 centipoise and its specific gravity is 0.88. Determine:
- (a) The absolute viscosity of the liquid(b) The kinematic viscosity of the liquid, in m2/s. (c) The shear stress on the upper plate. (d) The shear stress on the lower plate, in Pa.
- (e) The direction of each shear stress calculated in parts (c) and (d).

Classification of Non- Newtonian Fluid

Non-Newtonian Fluids $(\tau \neq \mu \frac{du}{dy})$

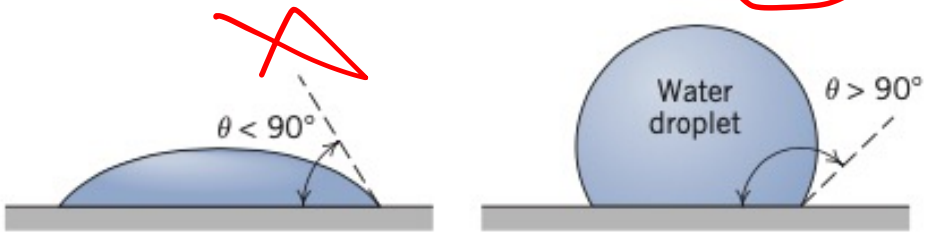
Purely Viscous Fluids		Visco-elastic Fluids
Time - Independent	**Time - Dependent**	
1. Pseudo plastic Fluids $\tau = \mu\left(\frac{du}{dy}\right)^n ; n<1$ Example: Blood, milk **2. Dilatant Fluids** $\tau = \mu\left(\frac{du}{dy}\right)^n ; n>1$ Example: Butter **3. Bingham or Ideal Plastic Fluid** $\tau = \tau_o + \mu\left(\frac{du}{dy}\right)^n$ Example: Water suspensions of clay and flyash	**1.Thixotropic Fluids** $\tau = \mu\left(\frac{du}{dy}\right)^n + f(t)$ *f(t)is decreasing* Example: Printer ink; crude oil **2. Rheopectic Fluids** $\tau = \mu\left(\frac{du}{dy}\right)^n + f(t)$ *f(t)is increasing* Example: Rare liquid solid suspension	**Visco- elastic Fluids** $\tau = \mu\frac{du}{dy} + \alpha E$ Example: Liquid-solid combinations in pipe flow.





Surface Tension

- Water droplets tend to appear somewhat flattened out.
- After waxing - you get a nice “beading” effect.
- Two cases shown in figure

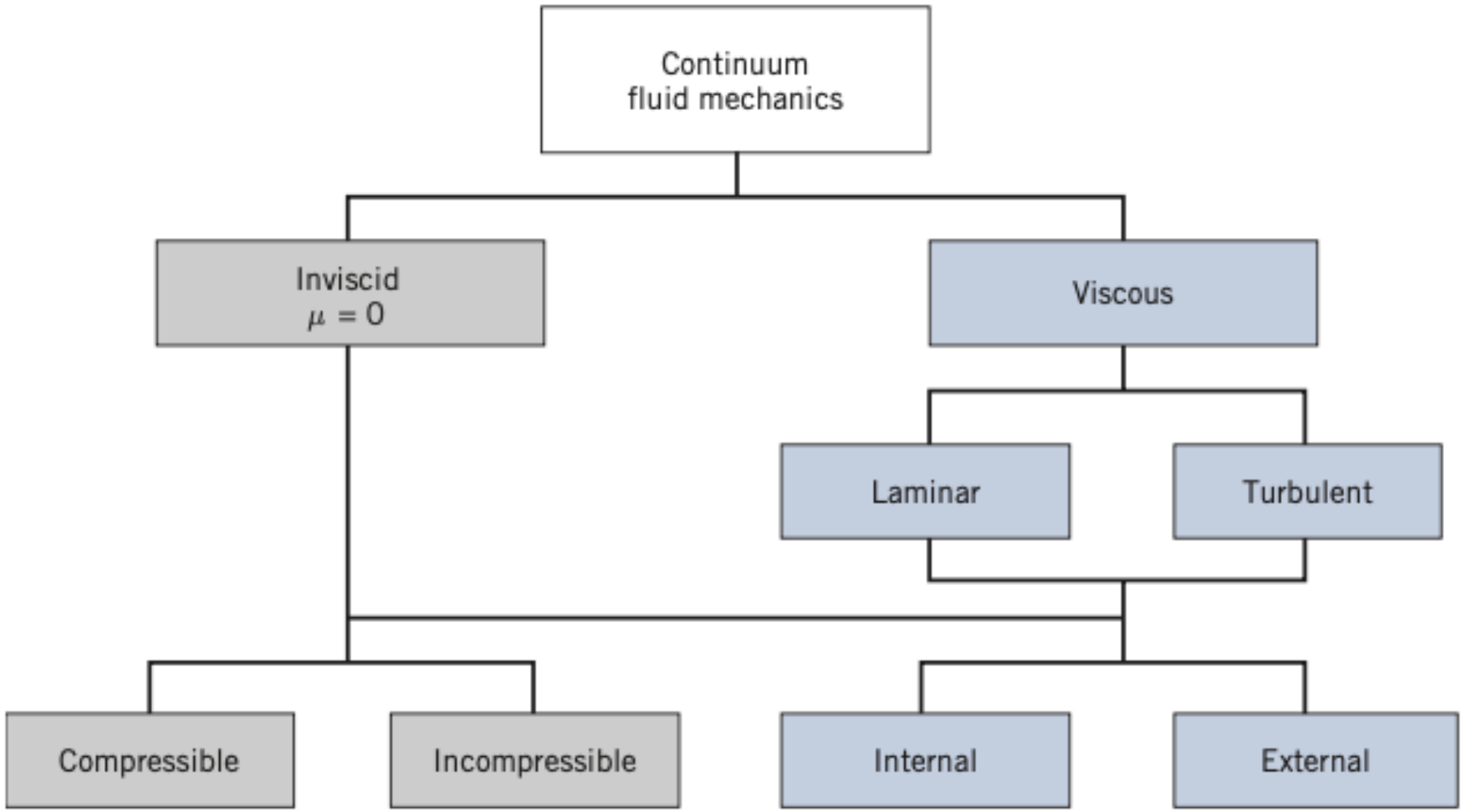


(a) A “wetted” surface (b) A nonwetted surface

- We define a liquid as “wetting” a surface when the contact angle θ , 90. and it is due to effects due to surface tension.
- Whenever a liquid is in contact with other liquids or gases, or in this case a gas/solid surface, an inter- face develops that acts like a stretched elastic membrane, creating surface tension.
- There are two features to this membrane: the contact angle, b, and the magnitude of the surface tension, (N/m). Which depend on the type or liquid and the type of solid surface (or other liquid or gas) with which it shares an interface.
- In the surface waxing, the contact angle changed from being smaller than 90°, to larger than 90°, because, in effect, the waxing changed the nature of the solid surface.
- The most important effect of surface tension is the creation of a curved meniscus that appears in manometers or barometers, leading to a (usually unwanted) capillary rise

- Calculate specific weight, density, and specific gravity of one litre of white kerosine which weighs 9 N.
- Calculate specific weight, density, and weight of one litre of white kerosine having specific gravity 0.8.
- Determine intensity of shear of an oil having viscosity = 1 poise. The ooil is used for lubricating the clearance between shaft of diameter 10 cm and bearing. The clearance is 1.5 mm and shaft rotate at 150 rpm.

Classification of Flow



- $Reynold\ Number$
- $= \frac{\rho \times V \times (D\ or\ \ l)}{\mu} = \frac{Inertia\ Force}{Viscous\ force}$
- Where ρ density of fluid
- V velocity of fluid
- D diamer if its pipe
- l characteristic length if its channel
- Re < 2000 then its Laminar flow
- Re > 2000< 4000 Transition flow
- Re > 4000 Turbulent flow

Fluid Statics

- **Fluid Statics**: It is the branch of fluid mechanics that deals with the behavior/response of fluid when they are at rest.

- **Pressure**, (average pressure intensity): It is the normal force exerted per unit area. It is denoted by P and is given by;

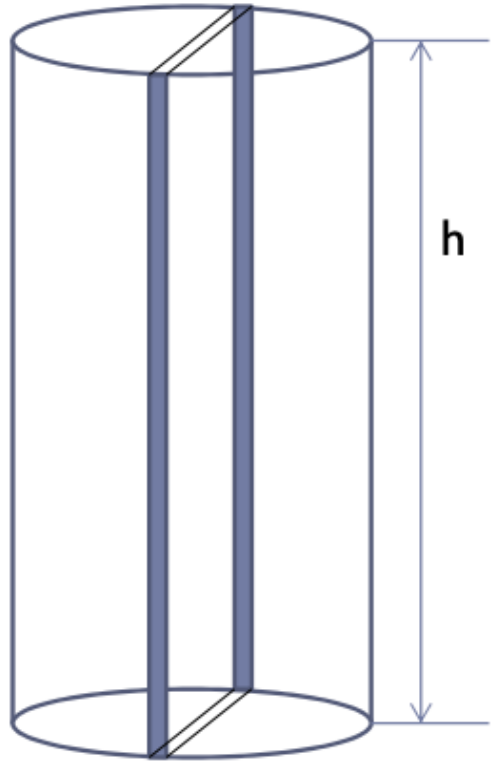
$$P = \frac{force}{area} = \frac{F}{A}$$

- **Units**
- **SI:** N/m^2 (called Pascal)
- **BG:** lb/ft^2 or lb/in^2 (called psi)
- **CGS:** $dyne/cm^2$

$$1\ bar = 10^5 N/m^2 = 10^5 Pascal$$

Pressure vs Water depth/height

- Consider a strip or column of a cylindrical fluid,



- *h= height or depth of strip of fluid*
- *γ= specific weight of fluid*
- *dA=cross-sectional area of strip*
- *dV=volume of strip*
- *dW=weight of strip*

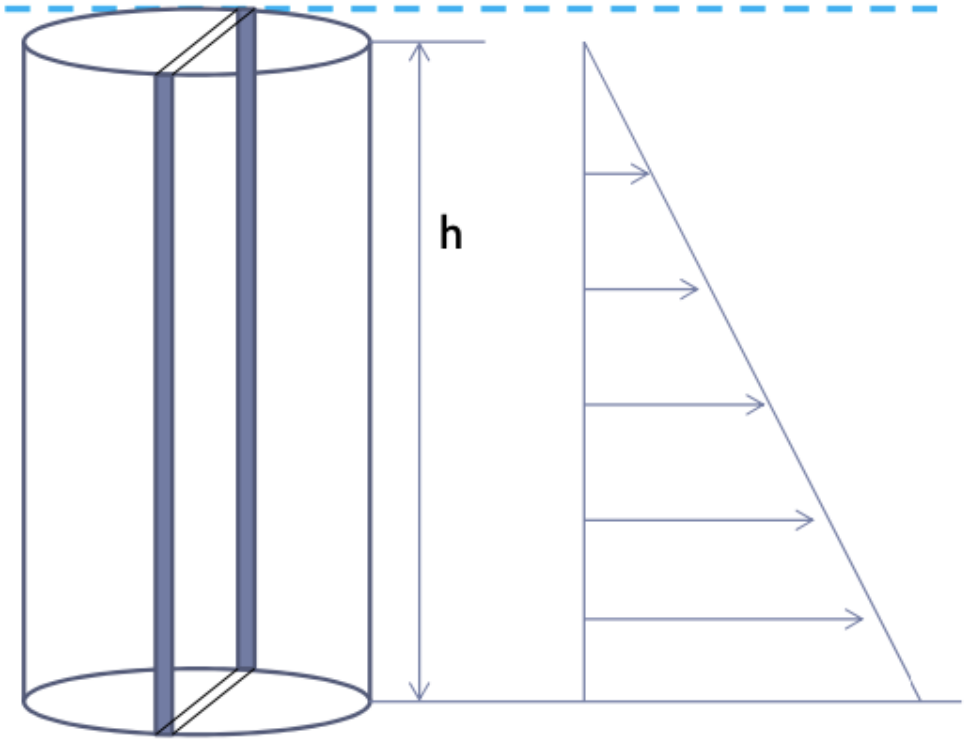
- Pressure at base of strip=dF/dA=dW/dA
- $P= \gamma dV/dA$
- $P= \gamma dA.h/dA$

$P=\gamma h$

Pressure vs Water depth/height

- $P=\gamma h$
- $P \propto h$

- For h=0, P=0
- For h=h, P=γh

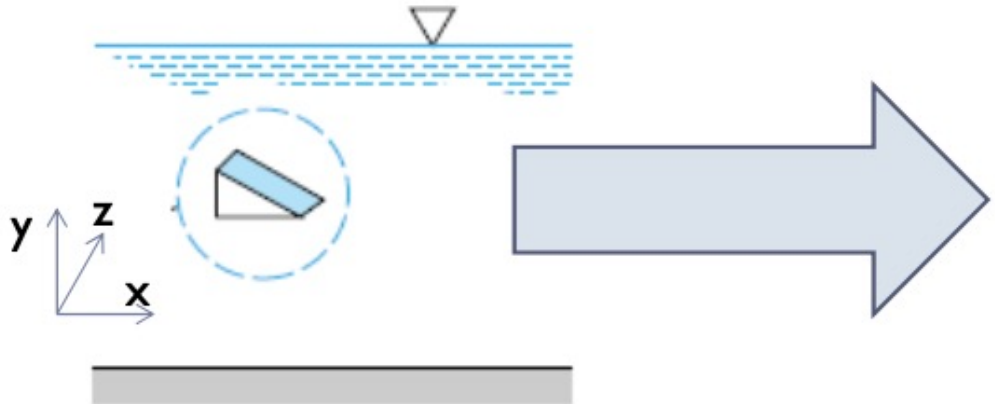


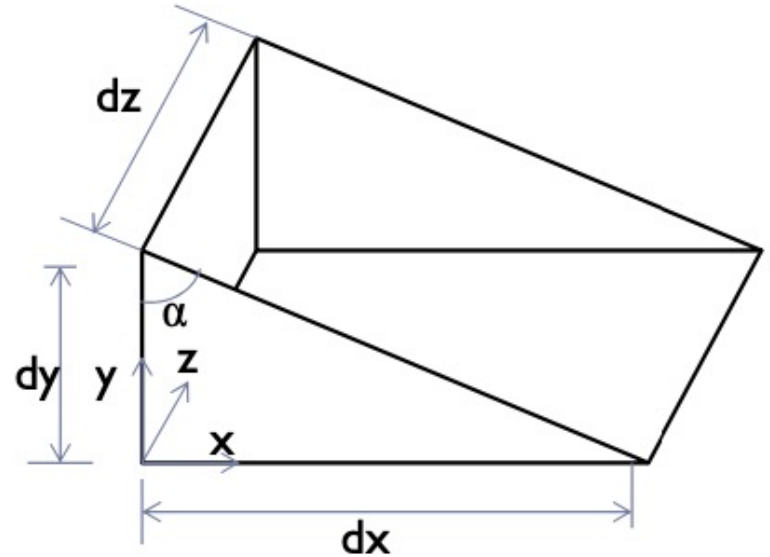
Pressure distribution diagram/pressure profile

- *As you know atmospheric pressure reduces, as we move to higher elevations. Is it because of h, as h reduces, P also reduces.*

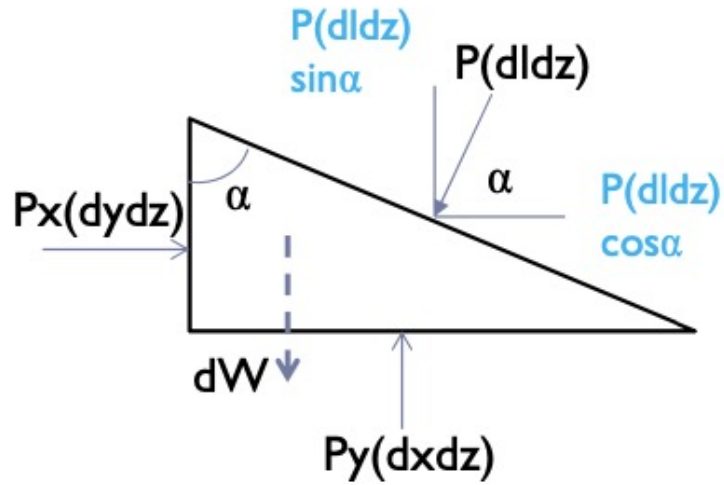
PASCAL'S LAW

- *"Pressure at any point in fluid is same in all directions when the fluid is at rest"*





- Consider a wedge shape element of fluid having dimension *dx, dy and dz* along *x, y and z* axis.
- dl= dimension of inclined plane making an angle α with the vertical
- *Px, Py, Pz* and *P* are pressure acting in *x, y, z* and perpendicular to inclined surface
- dW=weight of element



PASCAL'S LAW

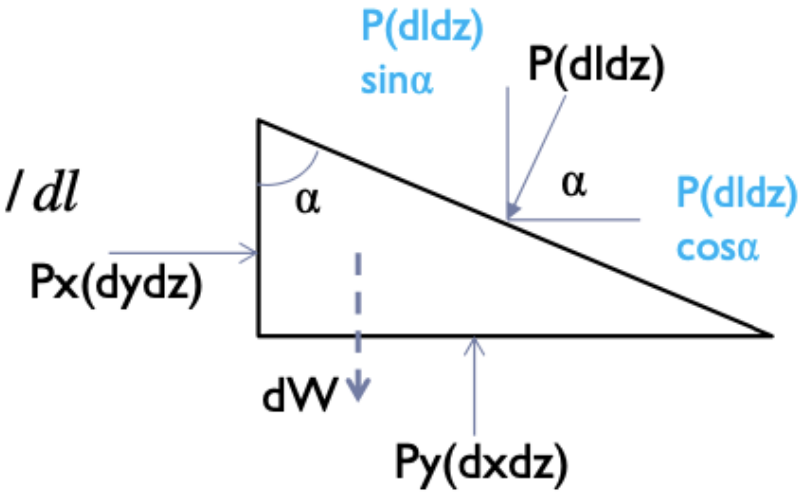
$$\sum F_x = 0$$

$$P_x dydz - P(dldz)\cos\alpha = o$$

$$P_x dydz - P(dldz)dy/dl = o \quad \because \cos\alpha = dy/dl$$

$$P_x dydz - P(dz)dy = o$$

$$P_x = P$$



$$\sum F_y = 0$$

$$P_y dxdz - dW - P(dldz)\sin\alpha = o$$

$$P_y dxdz - P(dldz)dx/dl = o \quad \because \sin\alpha = dx/dl \;\&\; dw \cong 0$$

$$P_y dxdz - P(dz)dx = o$$

$$P_y = P$$

PASCAL'S LAW

- Similarly by applying the conditions in z direction, it can be proved that

$$P_z = P$$

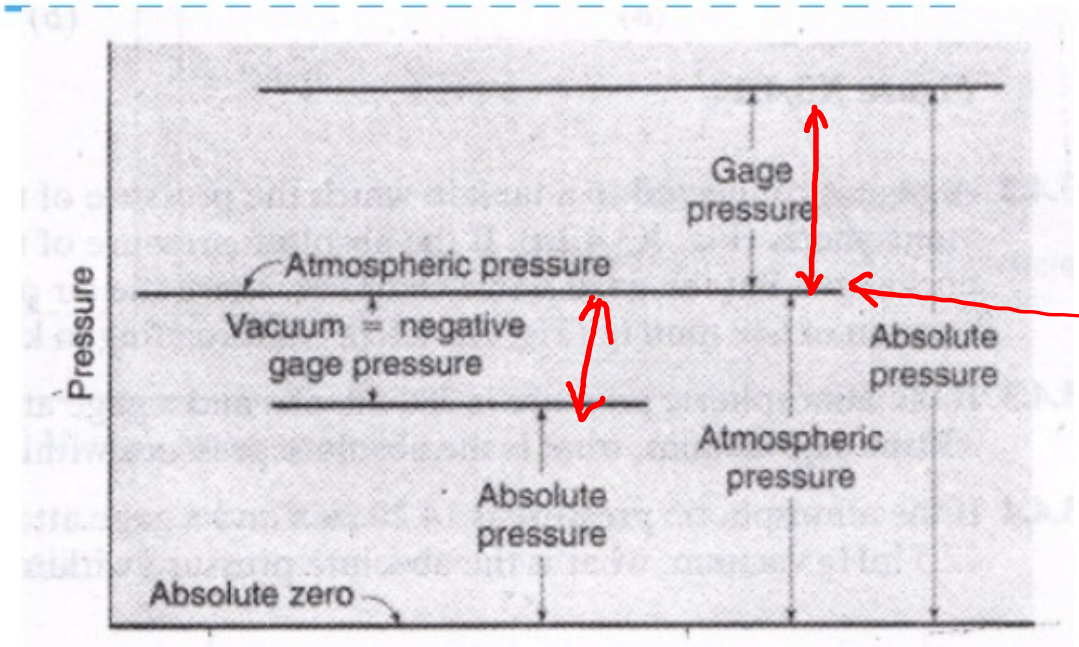
- Hence,

$$P_x = P_y = P_z = P$$

- The above states that the pressure acting on fluid particle is same in all directions when the fluid is at rest.

Absolute and Gauge Pressure

- Atmospheric pressure
- Gauge pressure
- Vacuum/negative pressure
- Absolute pressure



Pab

- **Atmospheric pressure:** Pressure exerted by atmosphere
- **Gauge pressure:** Pressure more than atmospheric pressure
- **Vacuum/negative pressure:** Pressure less than atmospheric pressure
- **Absolute pressure**: Pressure measure relative to absolute zero

$$P_{abs} = P_{atm} + P_g$$

$$P_{abs} = P_{atm} - P_{vac}$$

Absolute Pressure

- **Gauge Pressure (*Pg*):** It is the pressure measured relative to atmospheric pressure (P_{atm}) and is always above the atmospheric pressure
- It may be defined as normal compressive force per unit area

- **Vacuum Pressure** (P_{vac}): It is the pressure measured relative to atmospheric pressure and is less than the atmospheric pressure
- It may be defined as normal tensile force per unit area

- **Absolute Pressure**(P_{abs}): It is the pressure measured from absolute zero

$$P_{abs} = P_{atm} + P_g$$

$$P_{abs} = P_{atm} - P_{vac}$$

Atmospheric Pressure

- It is defined as weight of air per unit surface area of earth.

- It decreases with increase in elevation w.r.t. surface of earth.
- Standard atmospheric pressure at mean-sea-level is

=101.3KN/m^2

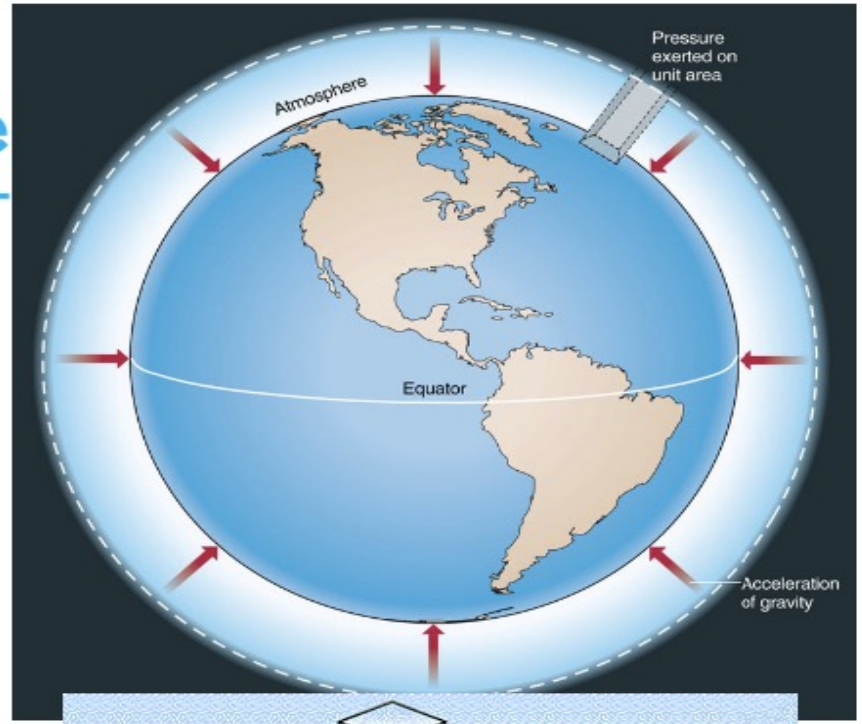
=1.013bar

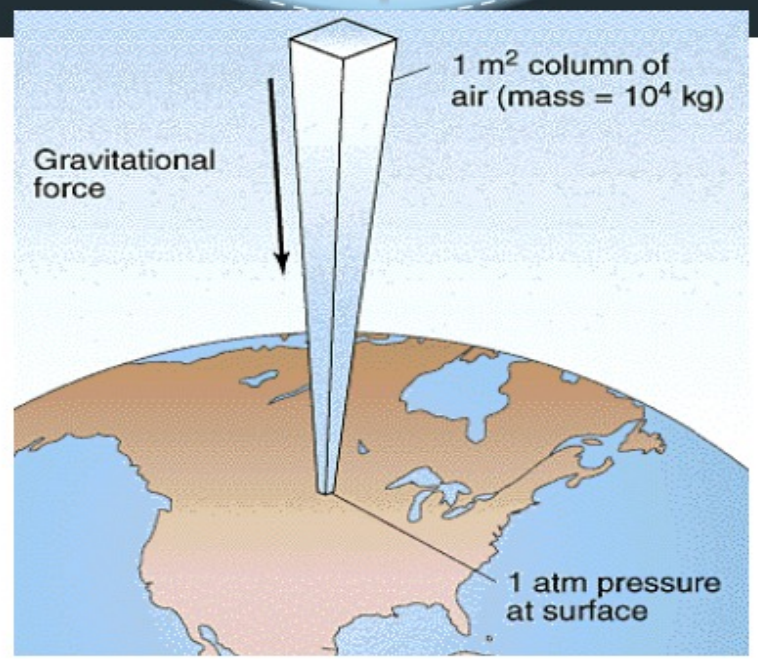
=14.7psi

=760mm of Hg

=33.9ft of water

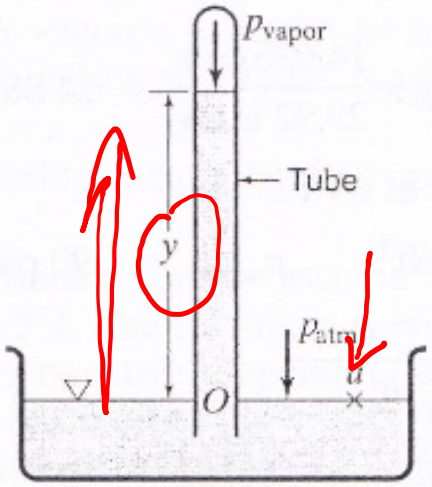
=10.3m of water



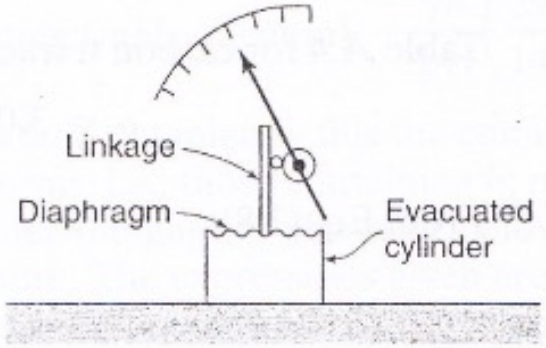


Measurement of Atmospheric Pressure

- **Barometer**: It is device used to measure the atmospheric pressure at any point on the earth.
- There are two types of barometer
- **(i) Liquid barometer**
- It measures the pressure with help of column of liquid
- **(ii) Aneroid barometer**
- It measures atmospheric pressure by its action on an elastic lid of evacuated box.



(a) Mercury barometer



(b) Aneroid barometer

Measurement of Pressure

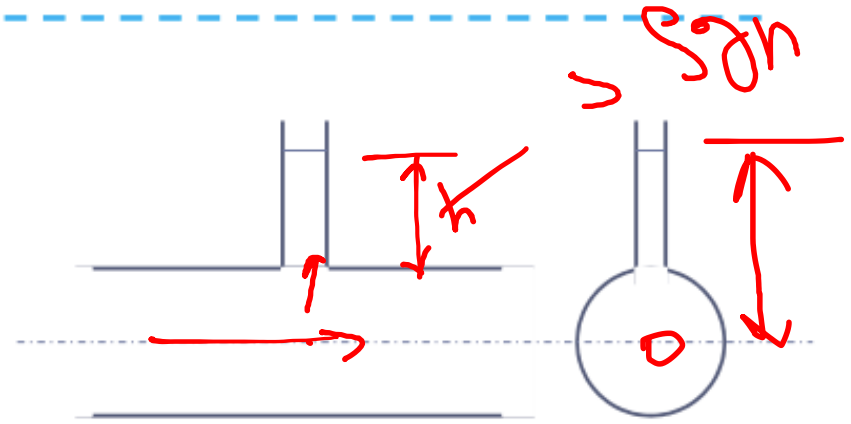
- The following devices are used for pressure measurement

- 1. Piezometer
- 2. Manometer
  - a) Simple manometer
  - B) Differential manometer
- 3. Mechanical Pressure Transducer (Bourden gauge)
- 4. Electrical Pressure Transducer

Not part of syllabus

1. Piezometer

- It is used to measure pressure in pipes or vessels.
- In it simplest form, it consists of a transparent tube open from other ends
- The diameter of tube should > ½" to avoid capillarity action
- Piezometers may be connected to sides or bottom of pipe to avoid eddies that are produced in the top region of pipe

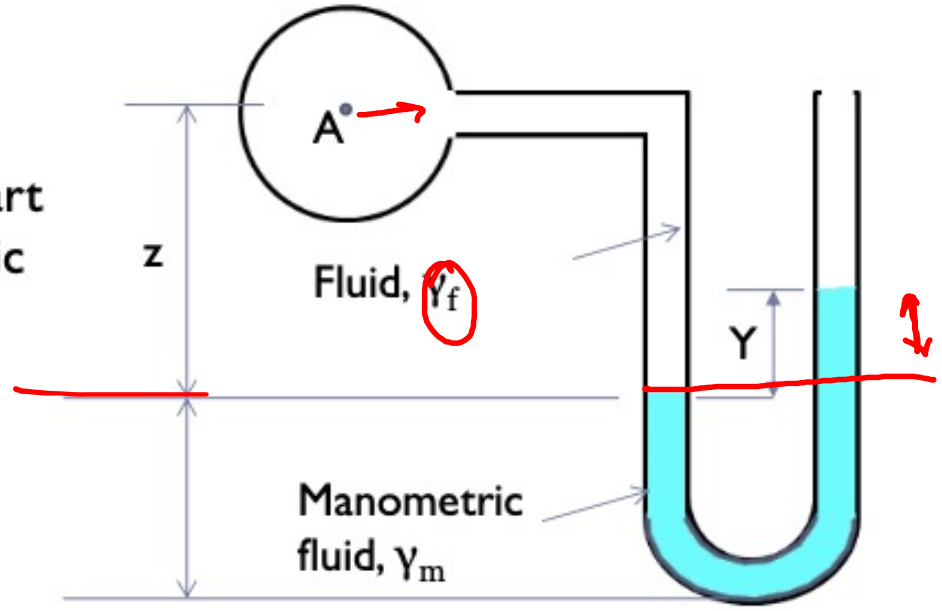


- When connected to pipes, the water level rises in it which gives a measure of pressure.

- **Limitations:**
  - It must only be used for liquids
  - It should not be used for high pressure
  - It cannot measure vacuum (-ve) pressure

2. Manometer

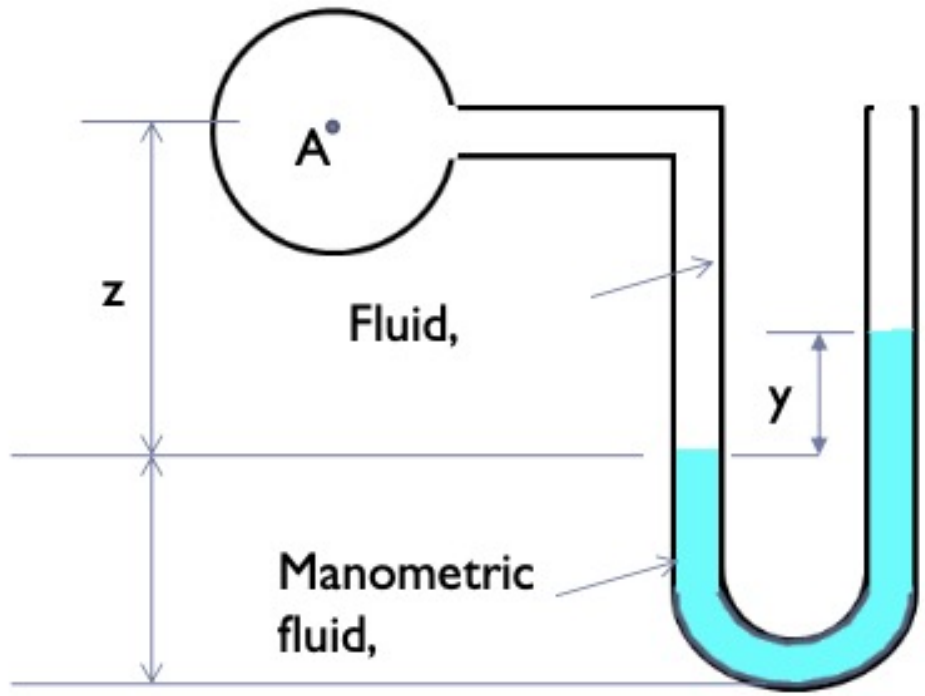
- **a). Simple Manometer**
- Figure shows a set up of simple manometer.
- It consists of a U shaped tube, part of which is filled with manometric fluid.
- One end of tube is connected with the pipe whose pressure is required to be determined.
- Due to pressure, level of manometric fluid rises on one side while it falls on other side.
- The difference in levels is measured to estimate the pressure.



- Y=Manometric reading
- γ_f =Specific weight of fluid in pipe
- γ_m =Specific weight of manometric fluid

2. Manometer

- **Manometric Fluids**
- 1. Mercury
- 2. Oils
- 3. Salt solution etc



- **Properties of manometric Fluid**
- 1. Manometric fluid should not be soluble/intermixale with fluid flowing in pipe whose pressure is required to be determined.

- 2. Lighter fluid should be used if more precision is required.

2. Manometer

- **Pressure measurement**

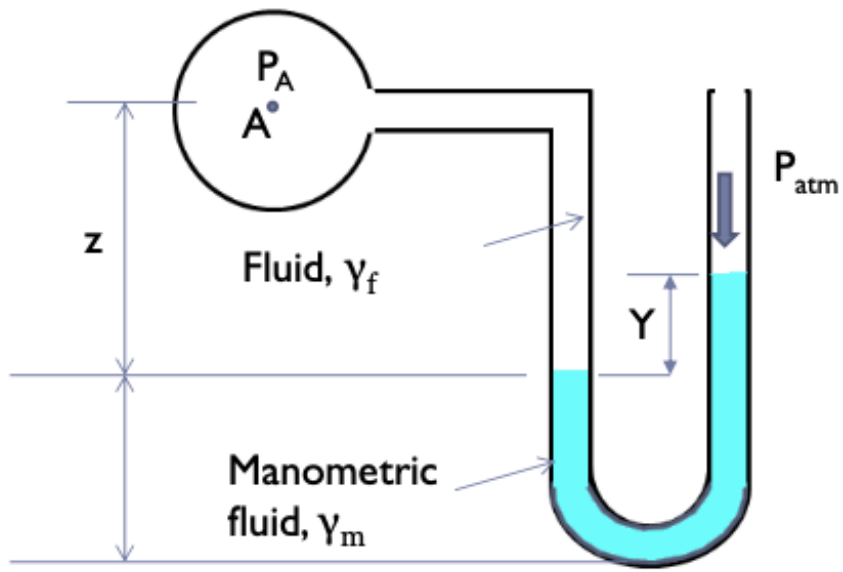
$$P_{Aabs} + \gamma_f Z - \gamma_m Y = P_{atm}$$

$$\because P_{Aabs} = P_{atm} + P_A$$

$$P_A + \gamma_f Z - \gamma_m Y = 0$$

$$P_A = \gamma_m Y - \gamma_f Z$$

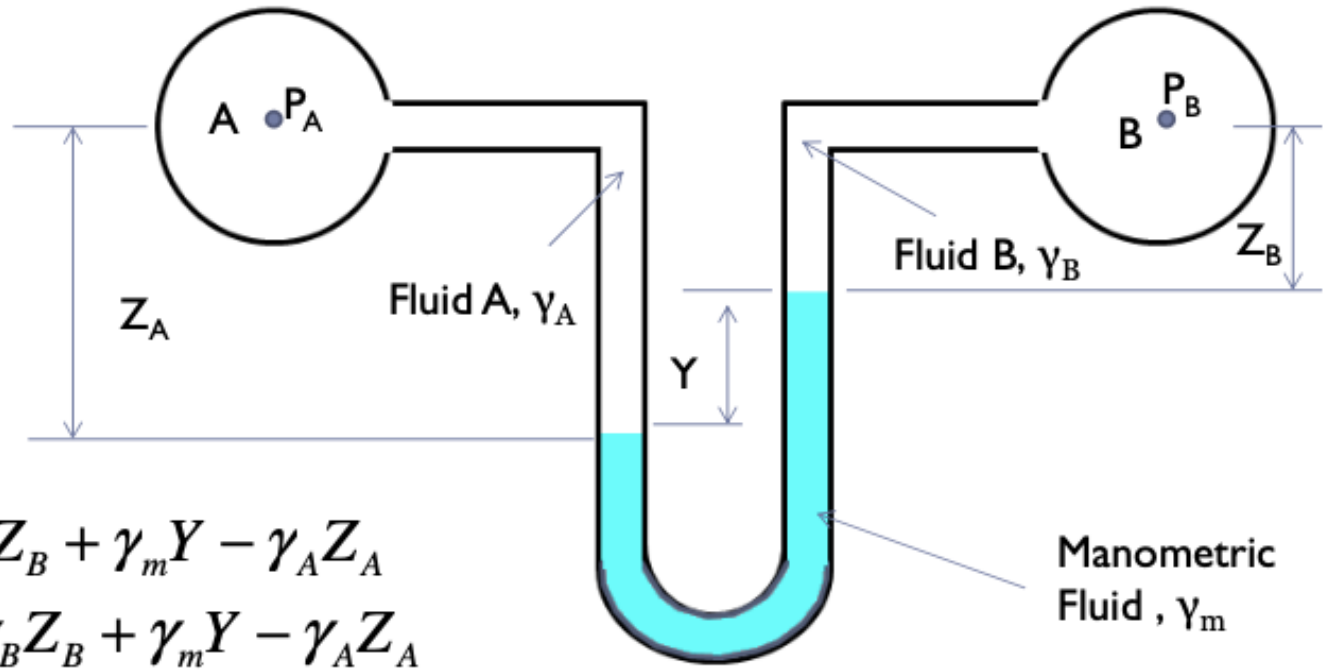
- **Above is a gauge pressure equation**



- **Sign Convention**
- **-ve: upward direction**
- **+ve: downward direction**

2. Manometer

- **b). Differential manometer**
- It is used to measure difference of pressure.
- **Case 1: when two vessels/pipes are at same level**

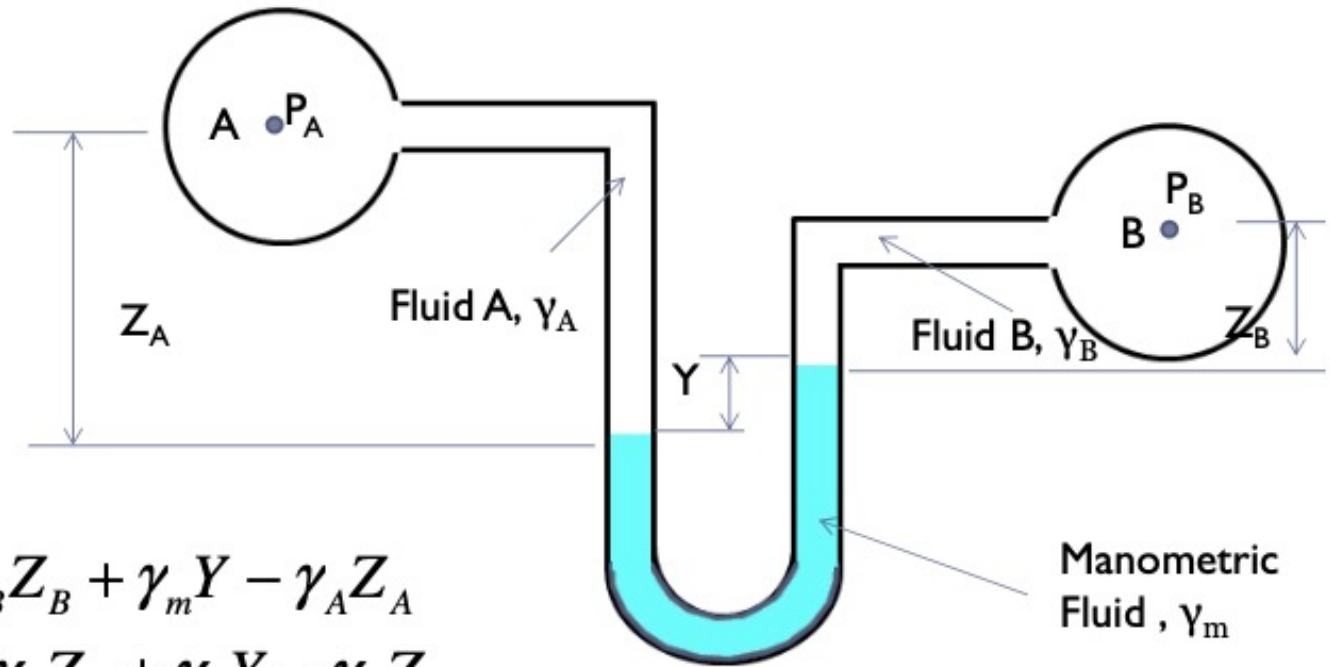


$$P_A = P_B + \gamma_B Z_B + \gamma_m Y - \gamma_A Z_A$$

$$P_A - P_B = +\gamma_B Z_B + \gamma_m Y - \gamma_A Z_A$$

2. Manometer

- **b). Differential Manometer**
- **Case 11: when two vessels/pipes are at different level**

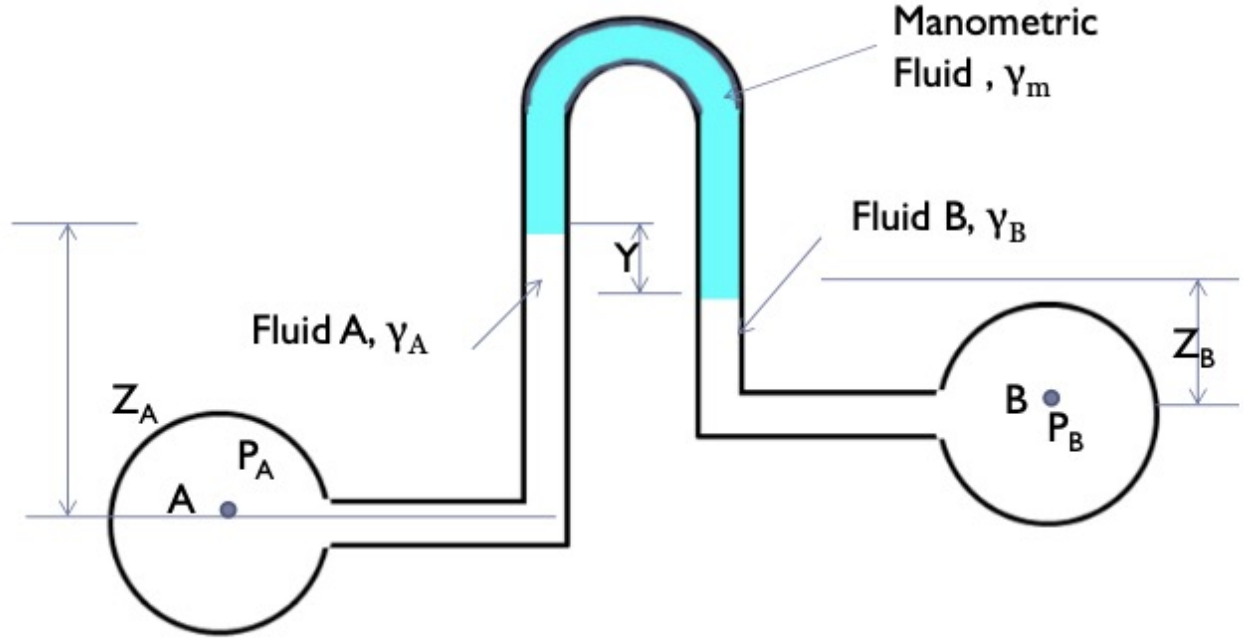


$$P_A = P_B + \gamma_B Z_B + \gamma_m Y - \gamma_A Z_A$$

$$P_A - P_B = +\gamma_B Z_B + \gamma_m Y - \gamma_A Z_A$$

2. Manometer

- **b). Differential manometer**
- **Case III: when manometer is inverted**



$$P_A = P_B - \gamma_B Z_B - \gamma_m Y + \gamma_A Z_A$$

$$P_A - P_B = -\gamma_B Z_B - \gamma_m Y + \gamma_A Z_A$$

Forces on Immersed Bodies

- **Hydrostatic Force:** It is the resultant force of pressure exerted by liquid at rest on any side of submerged body.

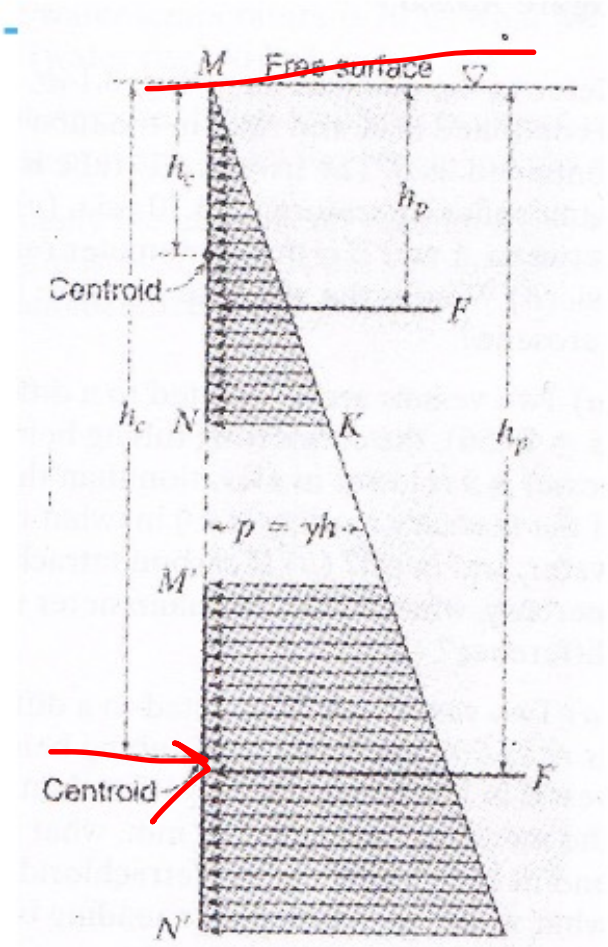
$$F = \int p dA = p \int dA = pA$$

- It is the summation of product of uniform pressures and elementary areas of submerged body

- It is equal to the product of submerged area and pressure at the centroid of the submerged area

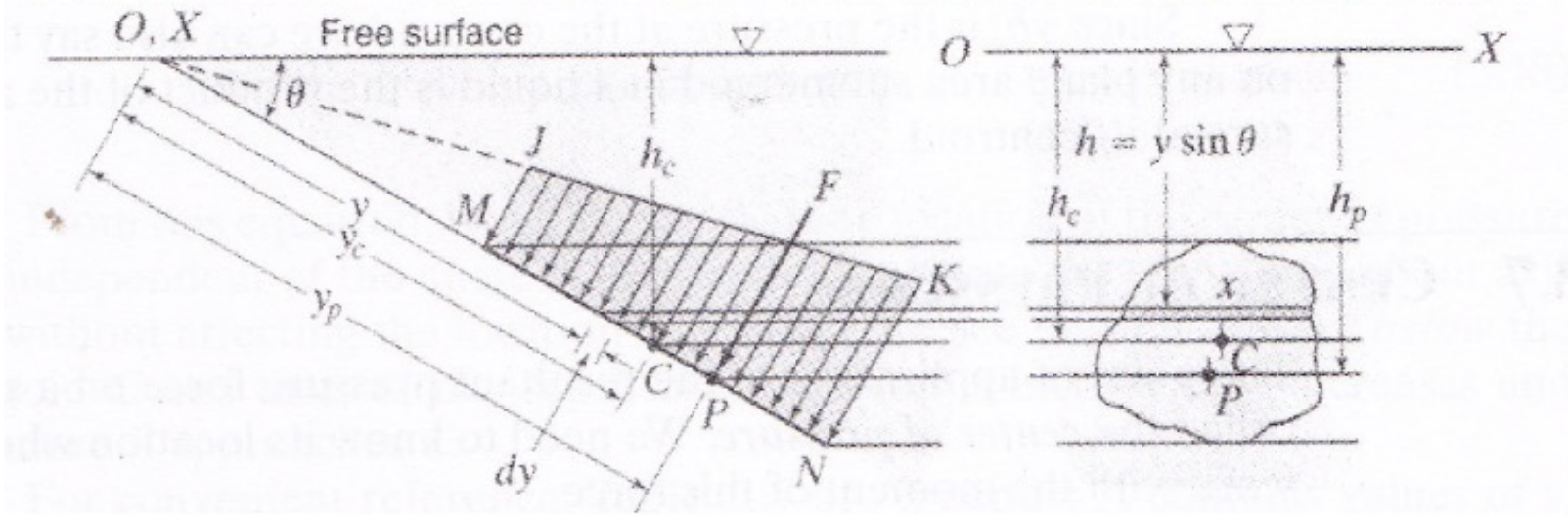
Forces on Plane Area

- **Center of pressure**
- The point of application of resultant force of pressure on a submerged area is called center of pressure.



Pressure distributions on two vertical plane areas (viewed from edges).

Forces on Plane Area



- A=total submerged area
- F=hydrostatic force
- θ=angle of submerged plane with free surface
- h_c=depth of center of area
- h_p=depth of center of pressure
- y_c=inclined depth to center of area
- y_p=inclined depth to center of pressure
- dA=elemtry area
- dF=force of pressure (hydrostatic force) on elementry area

$$dF = p(dA) = \gamma h(dA)$$

Forces on Plane Area

- Lets' choose an elementary area so that pressure over it is uniform. Such an element is horizontal strip, of width, x so $dA = xdy$. The pressure force, dF on the horizontal strip is

$$dF = p(dA) = \gamma h(dA)$$

$$p = \gamma h$$

$$h = y \sin \theta$$

- Integrating

$$\int dF = \int pdA = \int \gamma h dA = \int \gamma y \sin \theta dA = \gamma \sin \theta \int ydA$$

$$F = \gamma \sin \theta (y_c A)$$

$$y_c = \frac{\int ydA}{A}$$

- Where, y_c is the distance OX along the sloping plane to the centroid C of the area A. If h_c is the vertical depth to the centriod, then we have

$$F = \gamma h_c A$$

Forces on Plane Area

$$F = \gamma h_c A$$

- Thus, we find the total force on any plane area submerged in a liquid by multiplying the specific weight of the liquid by the product of the area and the depth of its centriod.

- The value of F is independent of the angle of inclination of the plane so long as the depth of its centroid is unchanged.

- Since γh_c is pressure at the centroid, we can also say that total pressure force on any plane area submerged is a liquid is the product of the area and the pressure at the centroid

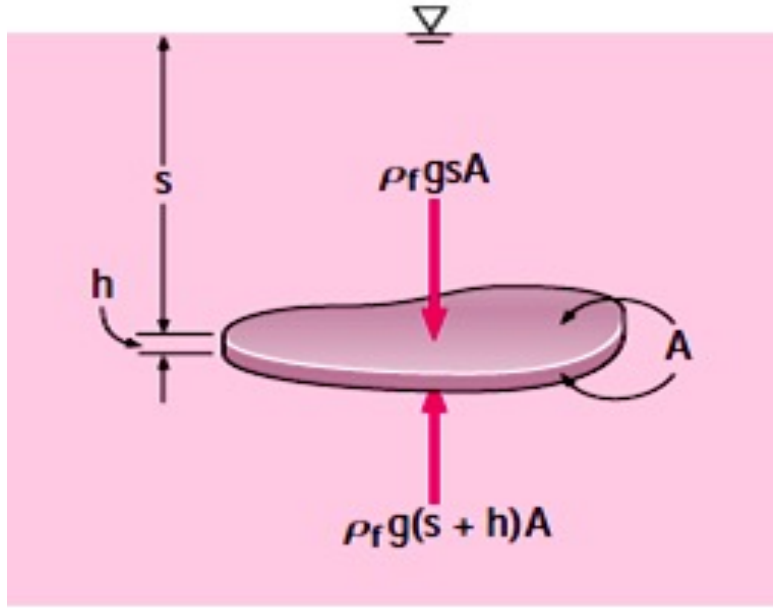
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Buoyancy

- It is a common experience that an object feels lighter and weighs less in a liquid than it does in air. This can be demonstrated easily by weighing a heavy object in water by a waterproof spring scale. Also, objects made of wood or other light materials float on water.
- These and other observations suggest that a fluid exerts an upward force on a body immersed in it. This force that tends to lift the body is called the **buoyant force** and is denoted by FB.
- The buoyant force is caused by the increase of pressure in a fluid with depth.
- If an object is immersed in a liquid, or floating on its surface, the net vertical force acting on it due to liquid pressure is termed buoyancy.

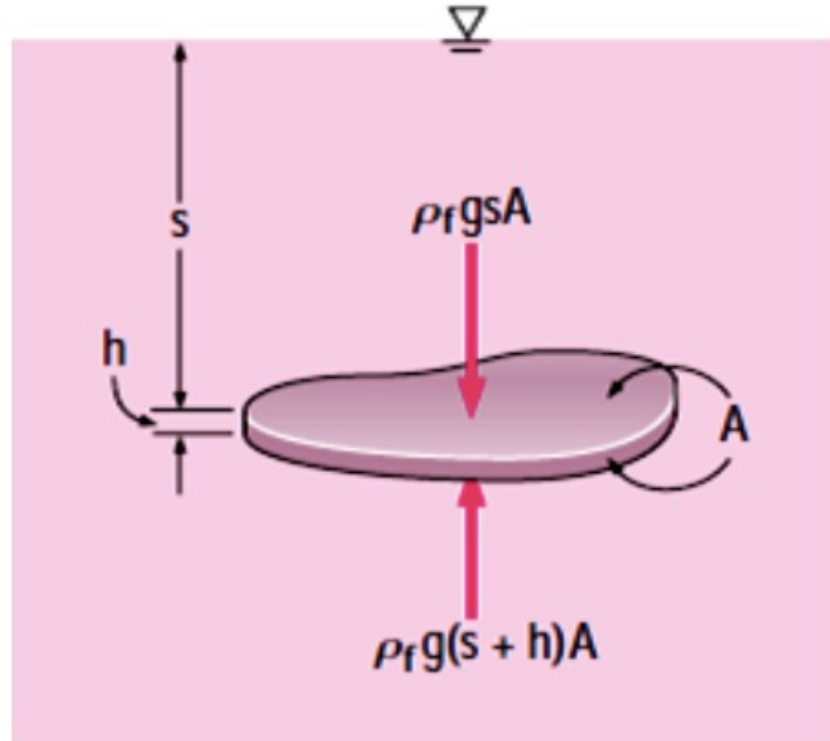
Buoyancy

- Consider, for example, a flat plate of thickness h submerged in a liquid of density ρf parallel to the free surface, as shown in the Fig.
- The area of the top (and also bottom) surface of the plate is A, and its distance to the free surface is s.
- The pressures at the top and bottom surfaces of the plate are **ρf gs** and **ρf g(s + h),** respectively.



Buoyancy, Floatation and stability

- Then the hydrostatic force $\mathbf{F_{top} = \rho_f gsA}$ acts downward on the top surface, and the larger force $\mathbf{F_{bottom} = \rho_f g(s + h)A}$ acts upward on the bottom surface of the plate.
- The difference between these two forces is a net upward force, which is the buoyant force,



$$F_B = F_{bottom} - F_{top} = \rho_f g(s + h)A - \rho_f gsA = \rho_f ghA = \rho_f gV$$

Buoyancy, Floatation and stability

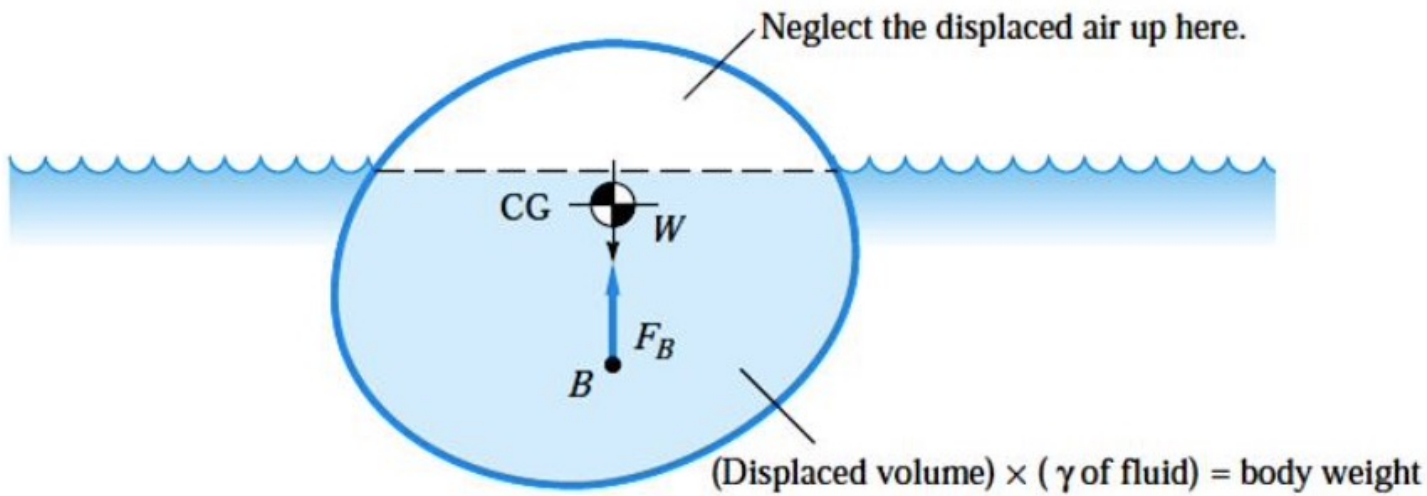
- where $\forall = hA$ is the volume of the plate. But the relation $\rho_f g\forall$ is simply the weight of the liquid whose volume is equal to the volume of the plate.
- Thus, we conclude that the buoyant force acting on the plate is equal to the weight of the liquid displaced by the plate.
- Note that the buoyant force is independent of the distance of the body from the free surface. It is also independent of the density of the solid body.
- This is known as **Archimedes' principle,** after the Greek mathematician Archimedes (287–212 BC), and is expressed as

> The buoyant force acting on a body immersed in a fluid is equal to the weight of the fluid displaced by the body, and it acts upward through the centroid of the displaced volume.

Buoyancy, Floatation and stability

- Floating bodies are a special case; only a portion of the body is submerged, with the remainder poking up out of the free surface.

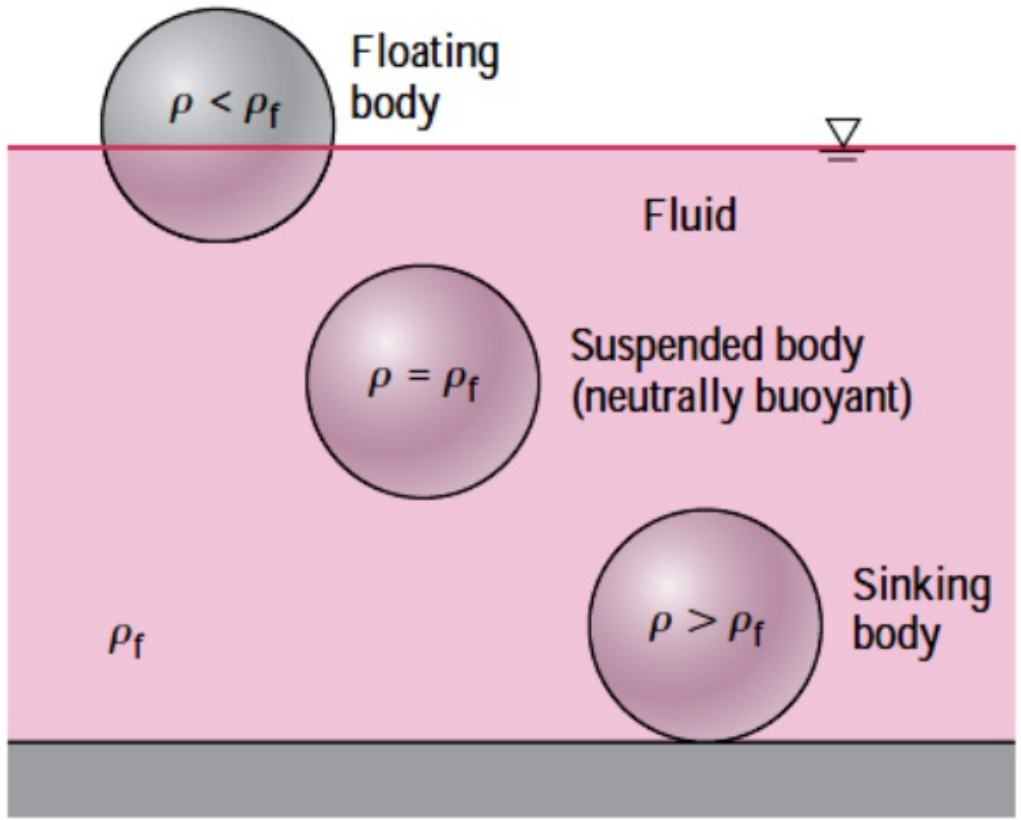
$$F_B = (\gamma)(\text{displaced volume}) = \text{floating-body weight}$$



Buoyancy, Floatation and stability

- A body immersed in a fluid

1) Remains at rest at any point in the fluid when its density is equal to the density of the fluid,
2) Sinks to the bottom when its density is greater than the density of the fluid, and
3) Rises to the surface of the fluid and floats when the density of the body is less than the density of the fluid



Buoyancy in Aviation

- Airships and balloons are termed “lighter-than-air” craft.
- The density of an ideal gas is proportional to molecular weight, so hydrogen and helium are less dense than air at the same temperature and pressure.
- Hydrogen is less dense than helium but extremely flammable, whereas helium is inert.
- Hydrogen has not been used commercially since the disastrous explosion of the German passenger airship Hindenburg in 1937

- A hot air balloon (approximated as a sphere of diameter 15 m) is to lift a basket load of 272 kg. To what temperature must the air be heated in order to achieve lift off?

